

2023

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : CC-12P

(OOPS Lab Using Java)

Full Marks : 30

Marks Distribution

Source Code	: 15
Output	: 05
Sessional	: 04
Viva-voce	: 06

SET-2

Answer *any one* question.

1. Create a super class 'Person' and one subclass 'Student'. Write a Java program to do the following :

(a) For 'Person' instance variables :

Name : String, address : String. Initiate variable through constructor, incorporate one method setperson() that updates Person variables, another method toString() that shows Person details as Person [name = ?, address = ?].

(b) For 'Student' sub class instance variables :

Program : String, year : String, fees : Double. Initiate both 'Student' and 'Person' variables through constructor, incorporate one method setStudent() that updates both 'Student' and 'Person' data, another method toString() that shows 'Person-Student' details as

"Person [name = ?, address = ?, program = ?, year = ? fees = ?]".

2. Write a Java program to find whether a region in the current string matches with a separate region in another string.

Sample output :

Str1 [0-7] == str2 [28-35]? true

Str1 [9-15] == str2 [9-15]? false.

3. Write a Java method to count all lines, words and characters of a file.
4. There is a class called Mypoint, which models a 2D point with X and Y co-ordinates. It contains :
- (a) Two instance variables x (int), y (int).
 - (b) A default constructor that construct a point at the default location (0, 0).
 - (c) An overloaded constructor that construct a point with the given x and y co-ordinates.
 - (d) A method getData() to take values of x and y from user.
 - (e) A method called linesegment (Mypoint m, Mypoint n) that finds out the gradient of the line segment and returns it from the function.
- Write the Mypoint class in Java and also write a class Gradient-check to test all the public methods defined in the class Mypoint.
5. Write a Java program to create and display a singly linked list. Also write a method to reverse the list.
6. Write a Java program to delete all consonants from an input string and print the resultant string.
7. Write a Java program to create a class student with following operations :
- (a) Create parameterized constructor to initialize the objects.
 - (b) Create a function isEqual() to check whether the two objects are equal or not which returns the Boolean value and gets two objects.
 - (c) Print the result in main method if objects are equal or not (take variables as your assumption).
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